

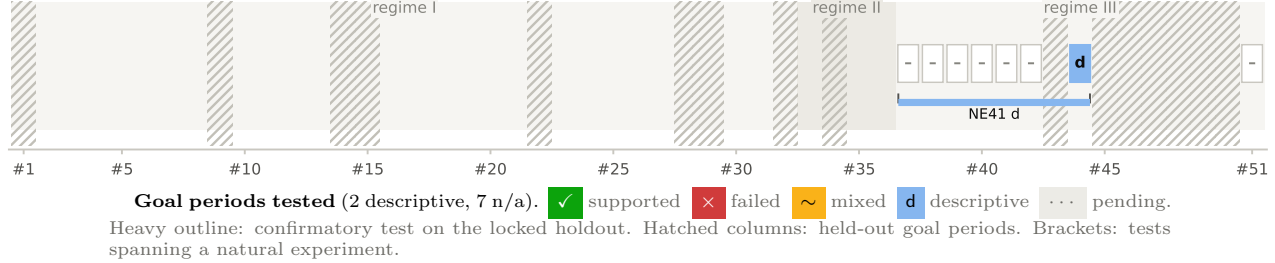
# H139: A fast read kick on a slow private well predicts how content variance splits between the two

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 45% Faithfulness 0.5/5 Usefulness 1/5

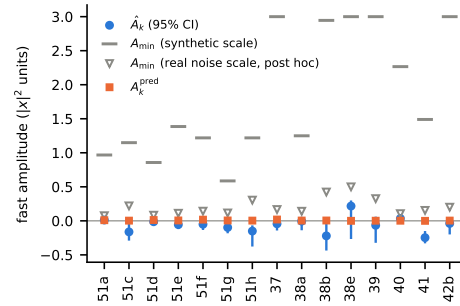
**Bottom line.** In 15 regime-III units the read-driven fast content variance predicted from H130's kick (median  $A_k^{\text{pred}}$  0.0044,  $\approx 2.5\%$  of the dynamic variance) lies 7 to  $>1,000\times$  below the smallest resolvable fast amplitude, so HH382's variance-split test is untestable at this data size.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



## WHAT HAPPENED

H130 found that in #51 a read kicks the reader's content ( $J_K = 0.044$ ) and the kick fades within  $\approx 7$  calls, while the agent's own well relaxes in  $\approx 100$  calls. HH382 asked whether that fast kick fixes, with no new fit, how much of an agent's content variance is fast. H139 measures the own content autocovariance at 1-call resolution in 15 regime-III units (#37–#42, #51) and compares its fast amplitude with the shot-noise prediction. A synthetic check on the real skeletons decided first whether the prediction is resolvable. It is not: the predicted fast amplitude is 7 to  $>1,000$  times below the smallest resolvable one in every unit. The verdict is inconclusive (below resolution).



**Observables.** Fitted fast amplitude  $\hat{A}_k$  (blue, 95% agent-day bootstrap CI) per testable unit, the predicted  $A_k^{\text{pred}}$  (orange) and the smallest resolvable amplitude  $A_{\min}$  (gray bar: synthetic; open triangle: rescaled to the real noise level, post hoc). The prediction sits far below both resolution marks in every unit.

## PHYSICAL MODEL

Two-rate Langevin relaxation (model 16) on the per-call clock. Content  $x = s + k$ : a slow part  $s$  held by the agent's private well (rate  $\gamma_s$ ) and a fast part  $k$  made of read kicks of size  $J$  that decay at  $\gamma_k$ . For independent kick directions, Campbell's theorem fixes the fast amplitude from the read rate  $\bar{r}$  (DQ1 ledger),  $J$  and  $\gamma_k$  (H130). The own autocovariance, corrected for the common drive, then has two exponentials and a plateau.

$$C(\tau) = A_k(1 - \gamma_k)^\tau + A_s(1 - \gamma_s)^\tau + B, \quad \tau \geq 1 \text{ call}$$

$$A_k^{\text{pred}} = \bar{r} J^2 / [1 - (1 - \gamma_k)^2], \quad Q_k = \hat{A}_k / A_k^{\text{pred}}$$

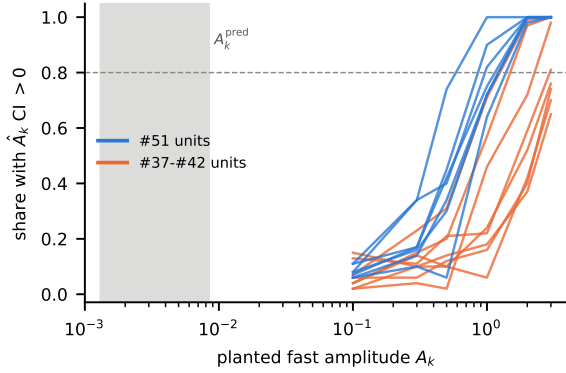
$$\text{S1: } A_{\min} > 2A_k^{\text{pred}} \Rightarrow \text{P1 untestable}$$

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, the read channel is not visible in an agent's overall content fluctuations: at about one read per call, H130's kick predicts a fast share of  $\approx 2.5\%$  of the dynamic content variance (median), well inside the noise of  $\sim 10^3$ – $10^4$  statement pairs per unit. To see what reading does to an agent, measure the response along the sender's direction (H130's dose design), not the agent's total variance.

## RESULTS VS PREDICTIONS

| Prediction (dated)                    | Observed                                            | Verdict        |
|---------------------------------------|-----------------------------------------------------|----------------|
| S1 $A_{\min} > 2A_k^{\text{pred}}$    | 15/15 units; ratio 92->2,000 (6.8-9,993 real scale) | as predicted   |
| S2 $Q_k$ unbiased $\pm 30\%$          | not resolvable at $\times 5, \times 20$             | failed         |
| S3 $R_{\text{fast}}$ separates worlds | 37-59% of replicates                                | failed         |
| P1 $Q_k \in [0.5, 2]$ (kill)          | $\hat{A}_k - 0.058$ $[-0.099, -0.017]$ vs 0.0044    | untestable     |
| P2 fast part $\propto$ reads          | not computed                                        | untestable     |
| P3 slow share $\geq 0.8$              | 13/15 units                                         | non-diagnostic |
| P4 slow share is an agent constant    | $\rho - 0.20, p 0.81$                               | failed         |
| P5 two rates beat one                 | 9/15 (null 34-58%)                                  | non-diagnostic |
| N1 NE41 fast part erased              | $R_{\text{slow}} 1.06$ $[0.98, 1.14]$               | untestable     |



**Second figure.** Synthetic check on the real skeletons of 15 units (real statements, calls, ledger reads, resets; 50-100 replicates per world). Share of replicates in which  $\hat{A}_k$ 's 95% CI excludes 0, against the planted fast amplitude. Blue: #51 units; orange: #37-#42 units. The gray band is the range of  $A_k^{\text{pred}}$ . Detection reaches 80% only at  $A_k \approx 0.6-3$ .

## GOAL PERIODS IN DETAIL

G44 d descriptive (no unit with  $\geq 3$  days).  
 NE41 d descriptive. N1 is untestable (S1 fired in every unit...  
 G37 - n/a (inconclusive: below resolution). S1 fired before...  
 G38 - n/a (inconclusive: below resolution). S1 fired before...  
 G39 - n/a (inconclusive: below resolution). S1 fired before...  
 G40 - n/a (inconclusive: below resolution). S1 fired before...  
 G41 - n/a (inconclusive: below resolution). S1 fired before...  
 G42 - n/a (inconclusive: below resolution). S1 fired before...  
 G51 - n/a (inconclusive: below resolution). S1 fired before...

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 0 |
| C adequacy             | 0 | H comparative     | 0 |
| D unfitted predictions | 0 | I transfer        | 0 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

The synthetic noise level was higher than the real one; rescaling to the real bootstrap SE (post hoc) still leaves  $A_{\min}/A_k^{\text{pred}} \geq 6.8$ . The drive correction subtracts conversational co-movement at short wall lags, and the lag-1 covariance sits below the lag-2 covariance in 14/15 units, so the constrained  $\hat{A}_k$  is negative (post hoc).  $\gamma_{\text{kick}}$  was not re-estimated in the shared weeks. Agent-level slow shares are unstable.

## ROUND 2

Measure the read echo along sender directions (H130's dose design), which has the power the variance split lacks. Explain the lag-1 dip (reply vs own-task alternation, or producing-call errors). Correct for the drive with unread partners only, and calibrate the synthetic noise to the real variance before new power claims.

- **Essence.** how much of an agent's moment-to-moment content is the echo of what it read.
- **H139-R1** Measure the echo where it is large.
- **H139-R2** The lag-1 dip.
- **H139-R3** Drive correction without conversation.
- **H139-R4** Calibrate the synthetic noise scale to the real  $-x-^2$  and dynamic variance before any new power claim.

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